Composed Image Retrieval: A Survey on Progress from CNN-based Fusion to CLIP and Beyond

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Abstract. Composed Image Retrieval (CIR) is an emerging task in computer vision that aims to retrieve a target image based on a reference image and a natural language modification. CIR bridges vision and language understanding, nudging models to reason compositionally and generalize beyond literal similarity. Today, we sense the impact of CIR employed almost everywhere, from Google Reverse Image Search to chatbots and E-Commerce recommendation systems. In this survey, we review around seven influential papers spanning from early models like TIRG, which proposed simple residual-based fusion techniques, to recent CLIP-based approaches such as Baldrati et al., moving to Zero-Shot CIR such as Pic2Word. We also discuss an attempt to address the non-differentiability of Recall@K, the most commonly used evaluation metric. Lastly, we explore training-free pipelines such as DualCIR and WeiMoCIR. Our analysis focuses on model architectures, loss functions, and evaluation techniques. We also highlight current limitations in the field, such as disjointed two-stage pipelines and the disconnect between training and evaluation objectives.

Keywords: Composed Image Retrieval, Conditioned Image Retrieval, Zero-Shot CIR, Training-Free CIR, Vision-Language Models, CLIP, Deep Metric Learning, Surrogate Loss, Retrieval Evaluation

1 Introduction

Composed Image Retrieval (CIR) is a relatively new but rapidly growing area in computer vision that combines image retrieval with natural language understanding. Unlike traditional content-based image retrieval (CBIR), which relies on a single visual query, CIR takes as input a **reference image** and a **textual modification** describing how the target image should differ from the input reference. This setup enables fine-grained retrieval use cases, such as finding *'the same person but in a red shirt'* or *'this chair but in wood instead of plastic'*.

In this survey, we trace the evolution from early fusion-based architectures to modern models built on pretrained vision-language representations such as CLIP. We then highlight the growing interest in zero-shot and training-free approaches. Along the way, we discuss core challenges in CIR: (i) the two-stage design of feature extraction and fusion, which limits end-to-end learnability; (ii) the disconnect between differentiable training losses and non-differentiable evaluation metrics such as Recall@K; and (iii) difficulty in handling rare or abstract modifications (e.g., transforming *'a dog on grass'* into *'a dog on the moon'*. The report analyzes seven representative papers spanning from 2019 to 2025.

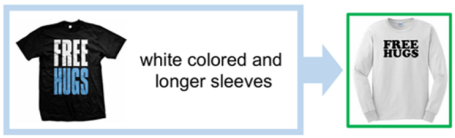

Fig. 1: An illustration of Composed Image Retrieval (CIR): the model retrieves a target image given a reference image and a natural language modification. Adapted from [9].

2 Literature Survey

2.1 TIRG: Text-Image Residual Gating

One of the earliest and most widely cited baselines in Composed Image Retrieval is the TIRG model [10]. TIRG introduced a novel approach to fusing vision and language by directly modifying the visual feature of a reference image using a textual modification. The core idea is to encode both the image and text into a shared space and apply a learned composition that alters the image feature based on the semantic information in the text.

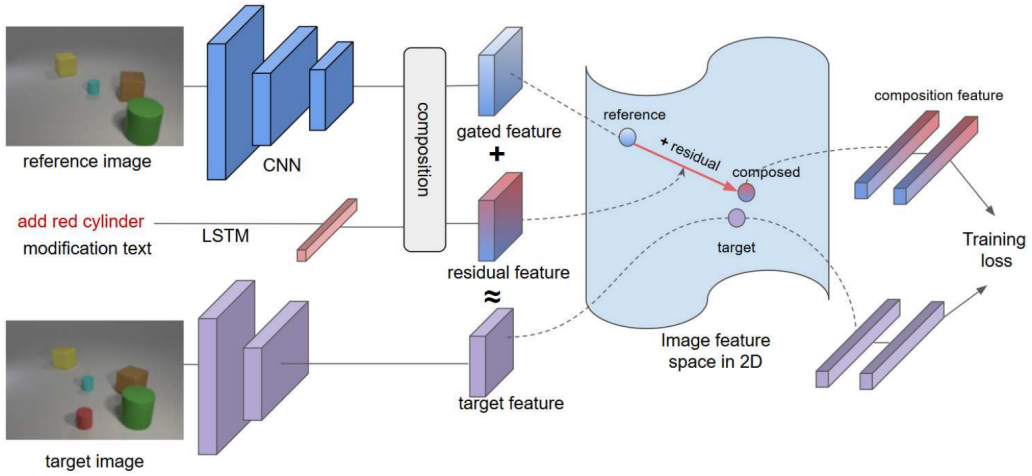

Fig. 2: TIRG system pipeline for training [10]

Model Architecture The model combines a **gating module** and a **residual module** to generate the composed query. Let $\phi_x \in \mathbb{R}^{W \times H \times 512}$ be the reference image feature map and $\phi_t \in \mathbb{R}^{512}$ be the encoded text, broadcast spatially. The **gating term**, $f_{\text{gate}} = \sigma(\phi_x * W_{g1} + \phi_t * W_{g2}) \odot \phi_x$, acts as a feature selector by suppressing or preserving image features based on text. Then **residual term**, $f_{\text{res}} = \text{ReLU}(\phi_x * W_{r1} + \phi_t)$, injects new, test-guided information. Finally, both terms are combined as $\phi_{xt}^{\text{rg}} = \omega_g f_{\text{gate}} + \omega_r f_{\text{res}}$ to encode the **composed** query for retrieval. $*$ denotes 2D convolution with 3×3 kernels W_{g1}, W_{g2}, W_{r1}, $\odot$ is an element-wise multiplication, and σ is the sigmoid function.

Strengths and Limitations TIRG is a simple and effective baseline. It offers interpretable fusion of visual and textual features through a residual-gating architecture. It performs well on structured datasets like *Fashion200k* [11] and *MIT-States* [5], and is easy to implement.

However, it struggles with abstract or rare modifications (e.g., *'replace suit with clown costume'*) due to limited compositional reasoning. Since TIRG is trained from scratch and cannot benefit from pretrained models like CLIP, its generalization to real-world queries remains limited.

2.2 CLIP: Contrastive Language–Image Pretraining

The introduction of CLIP by Radford et al. [6] marked a major paradigm shift in vision-language learning. Earlier models used to rely on manually annotated datasets such as ImageNet, but for CLIP it is different. CLIP was trained on a massive collection of 400 million pairs of (image, text) sourced from the Internet. Its training objective is simple: Given a batch of $\mathcal{N}$ image and text pairs, the model learns to **predict which caption belongs to which image** using a contrastive

loss. This is achieved by maximizing the cosine similarity of $\mathcal{N}$ correct pairs while minimizing the similarity of $\mathcal{N}(\mathcal{N}-1)$ incorrect pairs. This contrastive loss is defined as:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(I_i, T_i)/\tau)}{\sum_{j=1}^{N}\exp(\text{sim}(I_i, T_j)/\tau)} \tag{1}$$

Here, `sim(.,.)` is the cosine similarity between image and text embeddings, I_i and T_j are the i-th image and j-th text embeddings, and τ is a learnable temperature parameter. This approach results in a shared vision-language embedding space, where semantically similar image and text features are aligned.

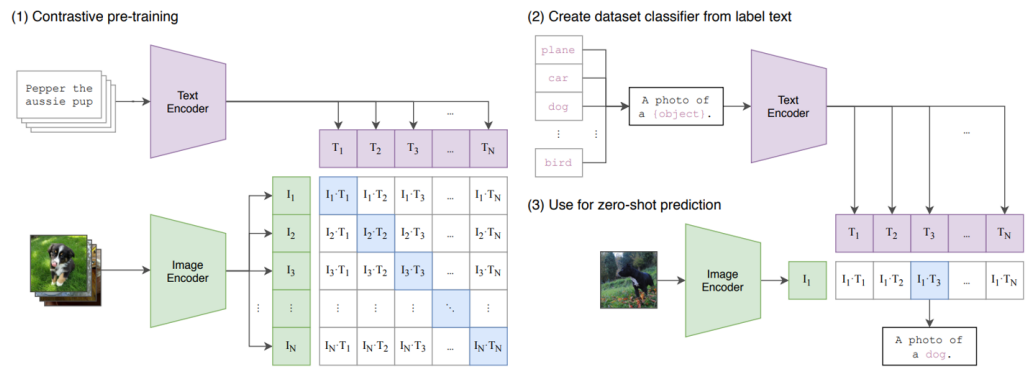

Fig. 3: Overview of the CLIP architecture. Separate image and text encoders map inputs into a shared embedding space. The model is trained via a contrastive loss to align matching image-text pairs. As noted in the original paper [8]: *"Predicting which text matches an image is easier and more scalable than predicting what the text says."*

2.3 CLIP-based Composed Image Retrieval

Baldrati et al. [3] adapt CLIP for CIR by keeping its pretrained encoders frozen or lightly fine-tuned, and introducing a lightweight fusion module **(Combiner)** to combine vision and language features as shown in Fig. 4. This balances CLIP's strong zero-shot transfer with learnable composition. It outperforms older baselines like TIRG on Fashion200k [11], MIT-States [5], and CIRR [6].

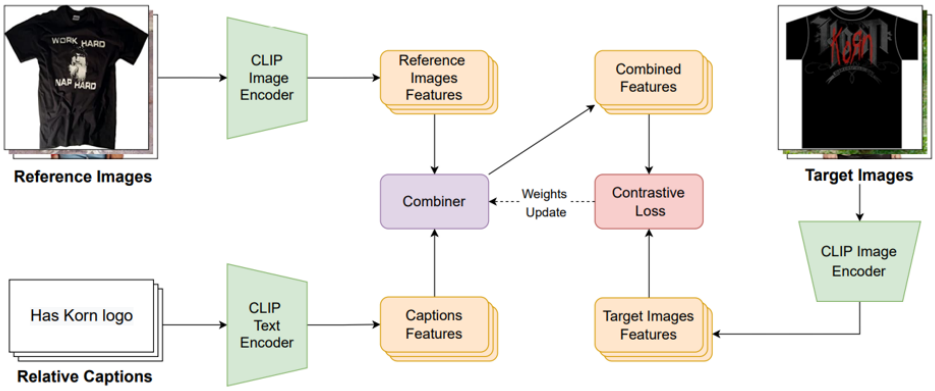

Fig. 4: Overview of Baldrati et al.'s CLIP-based CIR framework. A Combiner module fuses image and text features and is trained to align with the target image in CLIP space. [3]

The Combiner Network The model fuses CLIP image and text embeddings using a gated Multi-Layer Perceptron (MLP) to generate a composed feature vector:

$$f_{\text{combined}} = \sigma(g([\phi_x; \phi_t])) \cdot \phi_t + (1 - \sigma(g([\phi_x; \phi_t]))) \cdot \phi_x + h([\phi_x; \phi_t]) \tag{2}$$

where $\phi_x, \phi_t \in \mathbb{R}^d$ are the CLIP image and text embeddings, $[\phi_x; \phi_t] \in \mathbb{R}^{2d}$ is their concatenation, $g(\cdot), h(\cdot)$ are learned MLPs, and σ is the sigmoid function. The output f_{combined} is then ℓ_2-normalized: $f_{\text{combined}} \leftarrow \frac{f_{\text{combined}}}{\|f_{\text{combined}}\|}$.

Critique and Observations This method is simple, efficient, and highly modular and practical as it doesn't require finetuning large CLIP encoders. However, it uses a **two-stage pipeline** where feature composition and retrieval are decoupled. **Fusion in the post-CLIP** space prevents gradient propagation to the encoder.

2.4 Pic2Word: Zero-Shot Composed Image Retrieval

Pic2Word [9] addresses the scalability and generalization limits of traditional supervised CIR models (e.g., TIRG, Baldrati) that rely on costly triplet annotations. It enables zero-shot CIR by performing fusion entirely in the text domain, employing CLIP's pretrained encoders.

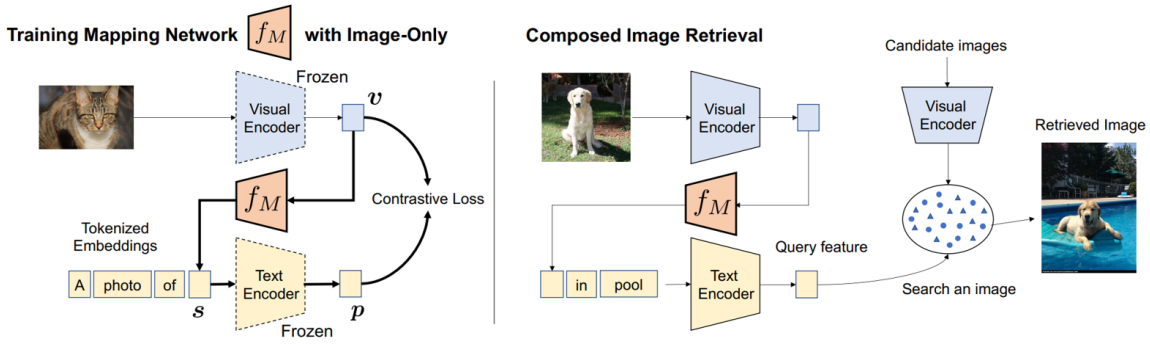

Fig. 5: **Left:** Pic2Word training architecture. The image is first projected into the language embedding space as a pseudo-word token, then composed with the modification text (e.g, `"a photo of"`). **Right:** During inference, the estimated token e.g., `"Golden Retriever Dog"` is used to fill in the given prompt sentence, e.g., `"in pool"`. (Figure Adapted from [9])

Table 1: Retrieval performance of representative CIR methods: TIRG on Fashion200K [4], Baldrati et al. on Fashion-IQ [11], and Pic2Word on CIRR [6]. Results are reported as in [10], [2], and [9].

Method	**R@1**	**R@10**	**R@50**
TIRG	14.10	42.50	63.80
Baldrati	—	43.78	66.93
Pic2Word	23.90	65.30	87.80

Model Architecture As illustrated in Fig. 5, the model first encodes an image x into a sequence of N **pseudo-word tokens**, where each token has a dimensionality of d. This is represented as the matrix as $s = \phi_{\text{img}}(x)W_{\text{proj}} \in \mathbb{R}^{N \times d}$. Here, ϕ_{img} is the CLIP image encoder, and W_{proj} prjects the image features into the language token space. This pseudo-word matrix s is then fused with a

relative caption or modification text t via the CLIP text encoder as $T([s;t]) = \phi_{\text{text}}(\text{concat}(s,t))$. Finally, they perform image retrieval by matching the composed vector $T([s;t])$ against candidate image embeddings $\phi_{\text{img}}(y)$ using cosine similarity.

Critique and Observations Pic2Word offers an elegant, supervision-free formulation for CIR. Its **early fusion** in the language space allows for strong generalization across complex compositions and unseen domains. However, mapping image features to pseudo-word tokens can lead to a **loss of fine-grained spatial details**, especially for complex scenes. Thus, while Pic2Word is impressive in zero-shot settings, it still may be outperformed by task-specific models trained with rich supervision.

2.5 An Attempt to Make Recall@K Trainable

As we have noticed so far, almost all CIR models are evaluated using Recall@K, but how is it defined exactly? It is the fraction of relevant images retrieved within the top-K ranked results. Unforunately, Recall@K is unsuitable as a loss function since it is non-differentiable. Existing models like TIRG, Baldrati, and Pic2Word relied on proxy losses (e.g., triplet loss, InfoNCE), which do not always align well with the Recall@K as the evaluation metric. Patel et al. [7] address this gap by proposing a differentiable surrogate function for Recall@K, enabling direct optimization via gradient descent.

Heaviside Definition For a query q, let P_q be the set of its relevant images, and $r_\Omega(q,x)$ the rank of image x among a database Ω. The Recall@K is defined as: $R_\Omega^K(q) = \frac{1}{|P_q|}\sum_{x\in P_q} H(K - r_\Omega(q,x))$, with the rank computed as: $r_\Omega(q,x) = 1 + \sum_{\substack{z\in\Omega \\ z\neq x}} H(s_{qz} - s_{qx})$

Here, s_{qx} is the similarity between query q and image x, and $H(\cdot)$ is the Heaviside step function. With this formulation, it is obvious now how the non-differentiability of H (its gradient is a Dirac delta) blocks backpropagation.

Differentiable Surrogate To make Recall@K trainable, the authors propose replacing Heaviside functions with sigmoids $\sigma_T(u) = \frac{1}{1+e^{-u/\tau}}$, controlled by temperature τ. By replacing Heaviside with sigmoid, and plugging the rank definition into the recall equation, the resulting surrogate becomes: $\widetilde{R}_\Omega^K(q) = \frac{1}{|P_q|}\sum_{x\in P_q} \sigma_{\tau_1}\left(K - 1 - \sum_{\substack{z\in\Omega \\ z\neq x}} \sigma_{\tau_2}(s_{qz} - s_{qx})\right)$. This approximation is fully differentiable, and the temperatures τ_1, τ_2 balance approximation quality and gradient density. Fig. 6 shows how sigmoid approximates Heaviside.

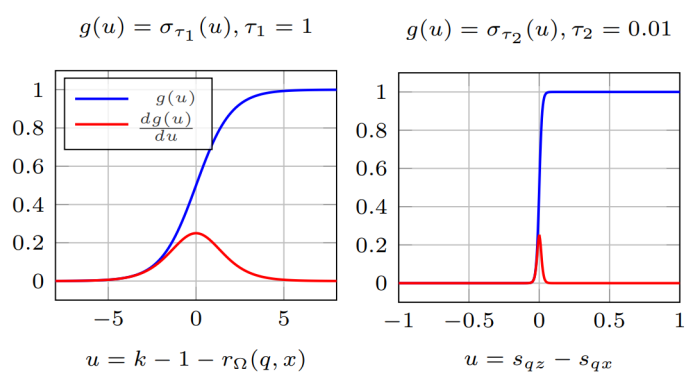

Fig. 6: Sigmoid approximation of the Heaviside step function with different temperatures. Lower temperatures produce sharper approximations with sparser gradient, while larger temperatures produce more lenient approximations with denser gradients.

2.6 DualCIR: Training-Free CIR via Dual-Directional Descriptions

From Training-Based to Training-Free Previous CIR models like TIRG, Baldrati, and Pic2Word require supervised or weakly supervised training, making them costly to adapt and limited in generalization. In contrast, Zhao et al. [13] propose **DualCIR**, a **training-free** approach that reframes CIR as text-to-image retrieval using only frozen pretrained components. A captioner, an LLM, and CLIP are all they need. No learning or fine-tuning is involved, retrieval is driven purely by descriptive rewriting.

Model Architecture Given a reference image I_r and a text modification T, a pretrained captioner ψ_c (e.g., BLIP VLM) generates a summary $c = \psi_c(I_r)$. A large language model then takes the caption c, the modifying text T, and a prompt p to produce three rewritten descriptions $\{t_e, t_p, t_n\} = \psi_R(c, T, p)$, where t_p is the positive description, it describes the target image content. t_n is the negative description, it paraphrases the reference image to mention objects which exist in the reference image but we want to avoid them in the target image. And t_e optionally captures the complete target description.

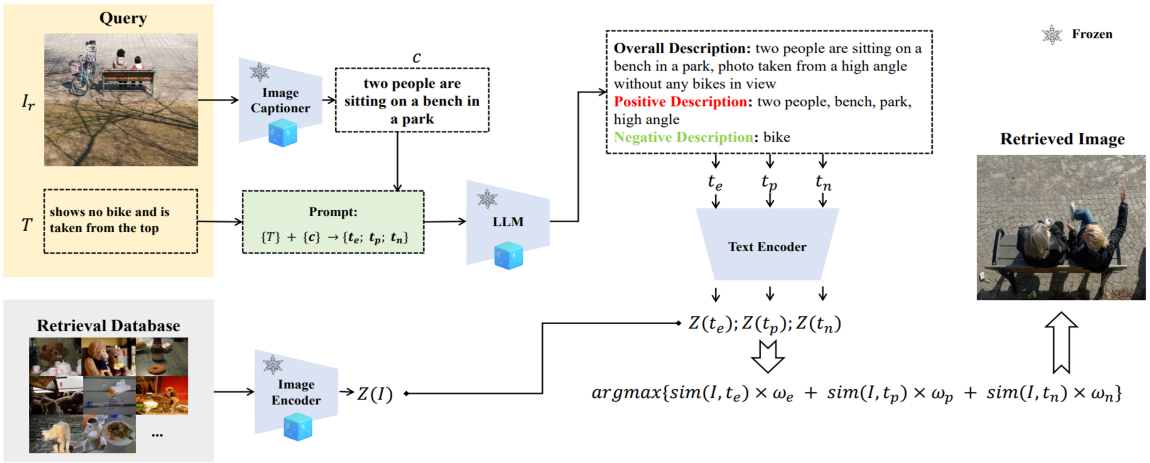

Fig. 7: **DualCIR Architecture.** Given a reference image I_r and a text modification T, the model uses a captioner (a VLM) and an LLM to produce dual-directional (positive and negative) textual prompts. These prompts are then encoded using **CLIP's text encoder**, and each image in the retrieval database is embedded via **CLIP's image encoder**. Then, cosine similarity scores are computed between the candidate image embedding $Z(I)$ and the text encoder embeddings $\{Z(t_e); Z(t_p); Z(t_n)\}$ and a weighted average is taken as the final score. The entire pipeline is training-free and uses only frozen pretrained models. [13]

2.7 WeiMoCIR: Training-Free Zero-Shot CIR via Weighted Modality Fusion

After we have seen how DualCIR came up with *training-free* CIR using LLMs and rewriting strategies, Wu et al. [12] propose an even **more minimalistic** alternative that they describe as **zero-shot training-free**. Their method, **WeiMoCIR** (Weighted Modality), avoids prompt engineering, rewriting, or any intermediate caption generation for the query. Instead, it performs CIR purely by fusing pretrained embeddings from images and text, relying on a simple but effective similarity fusion strategy.

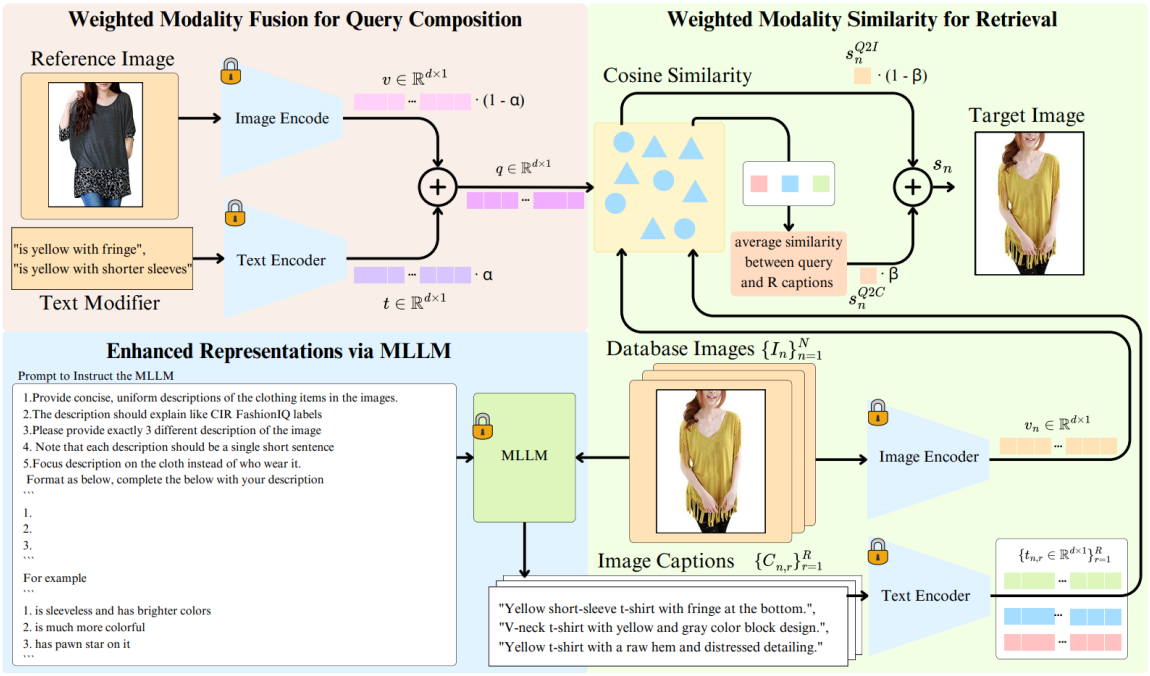

Fig. 8: **WeiMoCIR Pipeline.** The query q is formed as a weighted sum of the CLIP embedding of the reference image and the text, and a final similarity score s_n between the query q and database image I_n is computed as a weighted average of the query-to-image similairty S_n^{Q2I} and the query-to-caption similarity S_n^{Q2C}. Unlike DualCIR which captions the query, WeiMoCIR captions each database image with multiple captions. Finally, database images are sorted by their similarity scores with respect to the query, and the top K images are returned. [12]

Critique of Training-Free CIR DualCIR and WeiMoCIR showcase strong training-free CIR using only frozen models. DualCIR reframes retrieval as a text-to-image task via LLM-based rewriting, offering flexibility but relying on prompt quality. WeiMoCIR avoids rewriting, instead fuses query-to-image and query-to-caption similarities with fixed weights. It is simple and efficient, yet heuristic and less expressive. Both perform competitively, but still lag behind top-trained models in some fine-grained scenarios as shown in Table 2.

Table 2: **Comparison of Training-Based vs. Training-Free CIR Methods.** mAP@5 on CIRCO [1] and average Recall@10 on FashionIQ [11], as reported in Zhao et al. [13] and Wu et al. [12] respectively. DualCIR and WeiMoCIR narrow the performance gap, but training-based methods still sometimes achieve better results in some benchmarks.

Method	**Training?**	**mAP@5**	**Recall@10**
MagicLens	Yes (on 36.7M triplets)	**29.60**	30.70
Compodiff	Yes (on 18M triplets)	12.55	**36.02**
DualCIR [13]	No (only captioner + LLM)	27.86	30.37
WeiMo [12]	No (only MLLM)	—	31.54

3 Conclusion

This survey traced the evolution of Composed Image Retrieval (CIR) from early supervised fusion (TIRG), to CLIP-based fine-tuning (Baldrati), and to zero-shot methods such as Pic2Word. Then we looked at training-free pipelines like DualCIR, then WeiMoCIR pushed the envelope further by proposing a zero-shot training-free approach.

Many CIR pipelines followed two-staged designs, which limited end-to-end optimization. This motivated differentiable surrogates such as $RS@K$ aligned with ranking metrics like Recall@K.

Looking ahead, we believe that the key efforts are moving towards training objectives that optimize evaluation metrics directly, building richer datasets with multiple ground truths and interactive feedback; and most importantly, making training-free pipelines stronger and more adaptive. As on current benchmarks, it appears that purely training-based recipes are nearing saturation.

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